

Identify circle, its radius and diameter

Starter:

- The teacher will take a circular piece of paper and fold it so that a crease is formed and the circle is divided into two equal parts.
- Then the teacher will also draw a line across the crease explaining that this is the line segment of the circle and is called the diameter.
- Then the teacher will fold the circular strip two times, so that it is divided into four equal parts.

Look at the picture of a coin of Rs 5

It has a **circular** shape.

There is no side and no vertex in a **Circle**.



Activity

Radius

Now fold the circular strip two times. So, that it is divided in four equal parts.



Diameter

Take a circular piece of paper, fold it. So, that it is divided into two equal parts.

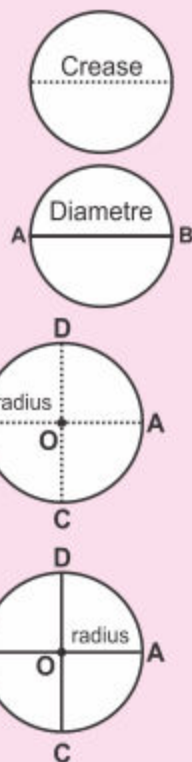
Draw a line in crease. This represents a line segment and such line segment is called **diameter** of the circle. Name the end points A and B and write as $\overline{AB}$. $\overline{AB}$ is diameter of the circle.

Now unfold and observe the crease. Join line and point of the creases. Name them as $\overline{AB}$ and $\overline{CD}$.

Two line segments $\overline{AB}$ and $\overline{CD}$ meet each other at one point (O). This point is called centre of the circle. The distance from this point O to boundary of the circle is called **radius**.

O is the centre of the circle and $\overline{OA}$ is its radius.

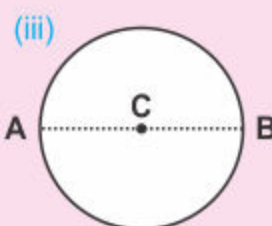
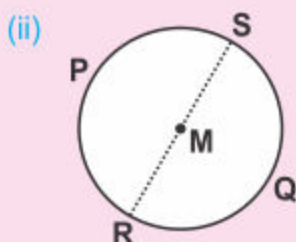
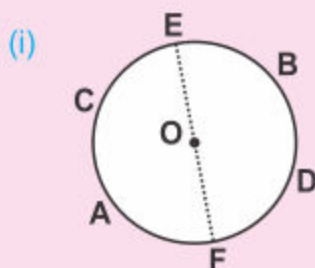
$\overline{AB}$ and $\overline{CD}$ are diameters of the circle.



Diameter of a circle is twice of its radius.

EXERCISE 55

(1) Name centre, radius and diameters in the following:



(i)
Centre

Diameter

Radius

(ii)
Centre

Diameter

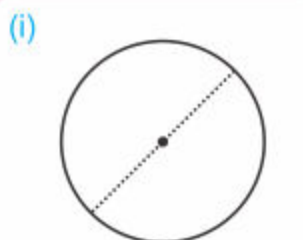
Radius

(iii)
Centre

Diameter

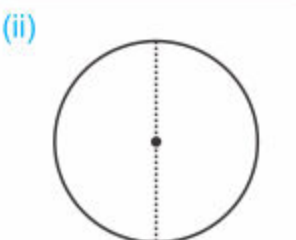
Radius

(2) Name diameters and radius of the following circles.



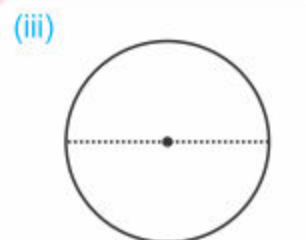
Diameter =

Radius =



Diameter =

Radius =



Diameter =

Radius =

Calculate perimeters of squares, rectangles and triangles.

Starter: The teacher will draw a triangle using the example given and explain what a perimeter is.

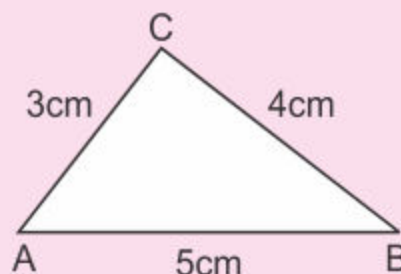
Perimeter

It is a triangle. It has three line segments.

Line segment AB is 5cm long;
line segment BC is 4 cm long and
line segment CA is 3 cm long.

The sum of these lengths is
 $5\text{cm} + 4\text{cm} + 3\text{cm} = 12\text{cm}$.

This sum is called **perimeter**.



The total length of all sides of any figure is called its **perimeter**.

Example 1: Find the perimeter of the rectangle.

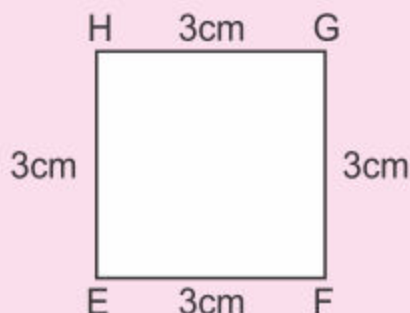
Solution:



$$\begin{array}{rcl}
 \text{Line segment AB} & = & 5\text{ cm} \\
 \text{Line segment BC} & = & 2\text{ cm} \\
 \text{Line segment CD} & = & 5\text{ cm} \\
 \text{Line segment DA} & = & + 2\text{ cm} \\
 \hline
 \text{Perimeter} & = & 14\text{ cm}
 \end{array}$$

Example 2: Find the perimeter of the square.

Solution:



$$\begin{array}{rcl}
 \text{Line segment EF} & = & 3\text{ cm} \\
 \text{Line segment FG} & = & 3\text{ cm} \\
 \text{Line segment GH} & = & 3\text{ cm} \\
 \text{Line segment HE} & = & + 3\text{ cm} \\
 \hline
 \text{Perimeter} & = & 12\text{ cm}
 \end{array}$$

Teacher's Note

Teacher should help the students to understand the concept of perimeter.

EXERCISE 56

Find the perimeter of following figures.



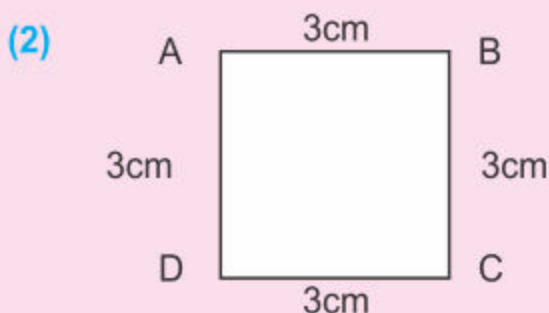
Line segment ZY =cm

Line segment YX =cm

Line segment XW =cm

Line segment WZ =cm

Perimeter =cm



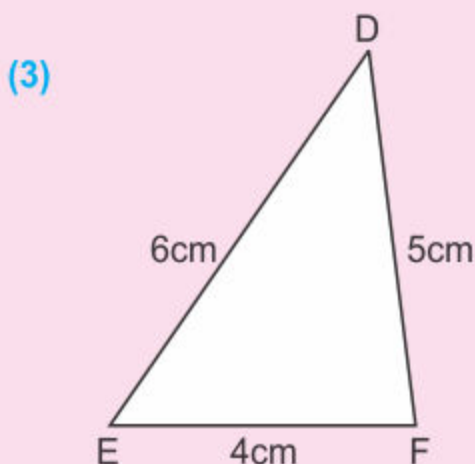
Line segment AB =cm

Line segment BC =cm

Line segment CD =cm

Line segment DA =cm

Perimeter =cm



Line segment EF =cm

Line segment FD =cm

Line segment DE =cm

Perimeter =cm

- (4) A Square shape table has each side 38 cm. Find perimeter of surface of the table.

- (5) The length and breadth of a window is 75 cm and 50 cm respectively. Find its perimeter.

7.1 PICTURE GRAPH

Read and interpret a picture graph

On the board the teacher will draw the pictures given and ask the children what information is shown in these pictures. After discussing a few responses, the teacher will explain that the same information can be shown even more briefly by counting in 2's.

The teacher will show this by drawing an appropriate diagram on the board.

Look at the picture.

First row	
Second row	

There are 10 boys in the first row and 6 girls in the second row. This information can be shown in picture as follows.

First row	
Second row	

Here we can tell the number of boys or girls simply by counting in 2's. The same information can be shown briefly as follow.

Here  represents 2 boys . So, there are 10 boys in first row.
and  represents 2 girls . So, there are 6 girls in second row.

This is picture graph.

Teacher's Note

Teacher should help the students understand the reading of a Picture graph by using other symbols.



Activity

This picture graph represents the number of different animals in a farm.

Cow	Goat	Horse	Camel
			
			
			
			

One  represents 2 cows.	One  represents 5 goats.
One  represents 2 horses.	One  represents 5 camels.

Read the picture graph and answer the following:

- (1) How many cows are there in the farm? $4 \times 2 =$
- (2) How many goats are there in the farm? $6 \times 5 =$
- (3) How many horses are there in the farm? $5 \times 2 =$
- (4) How many camels are there in the farm? $4 \times 5 =$
- (5) What is the total number of animals?

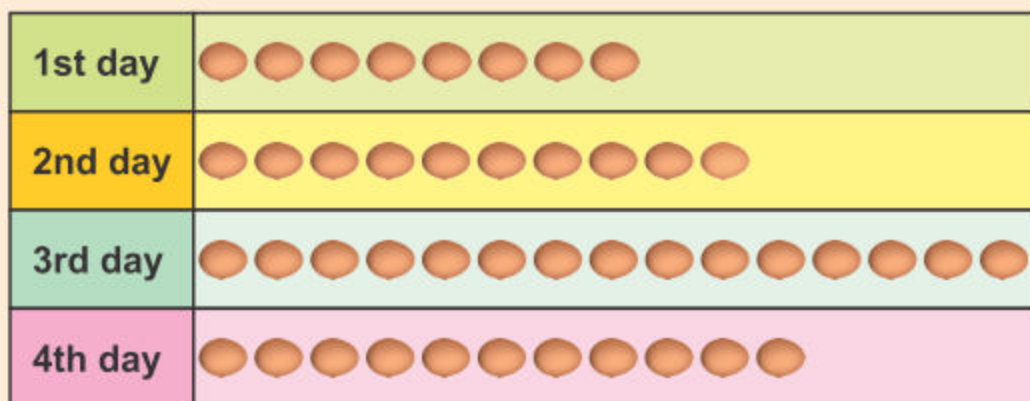
Teacher's Note

Teacher should help the students for solving the activity.

EXERCISE 57

- (A) The following picture graph shows the number of eggs sold by a shopkeeper in four days.

One  represents 5 eggs.

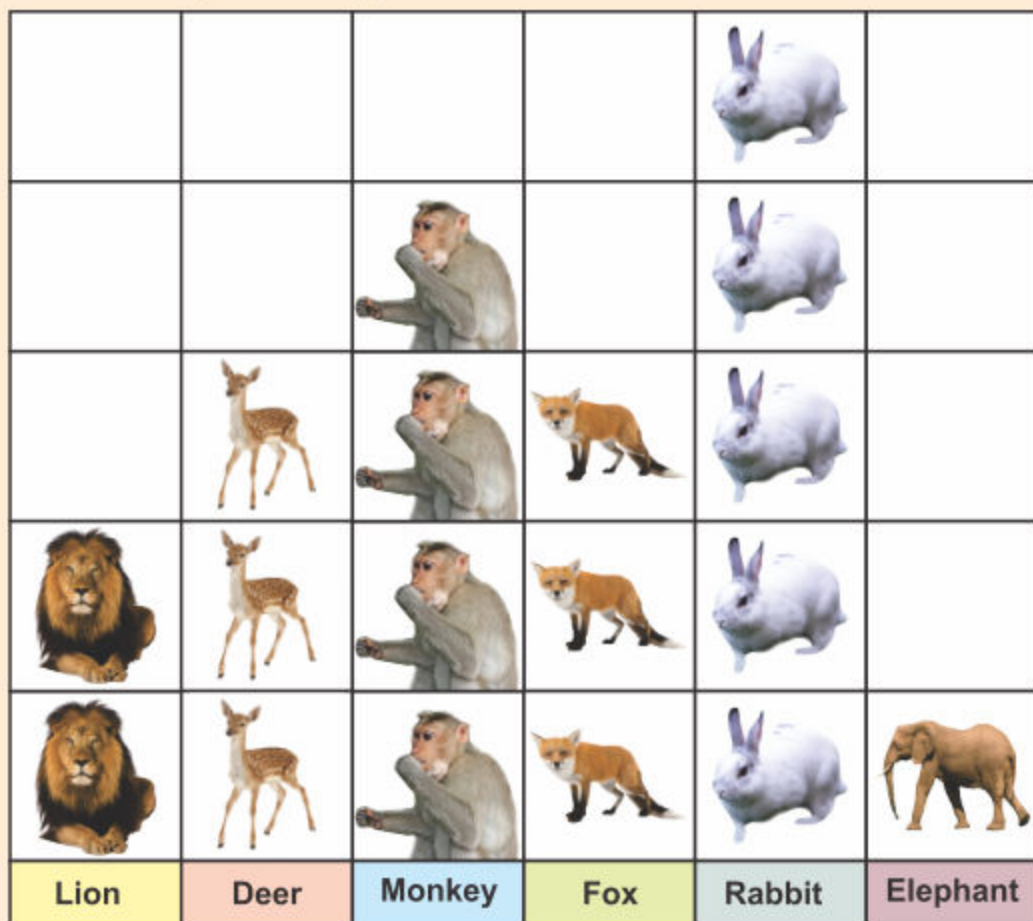


Read the above graph and answer the following questions:

- (1) How many eggs are sold on the last day?
- (2) On which day the least number of eggs are sold?
- (3) On which day the highest number of eggs are sold?
- (4) How many eggs are sold in first two days?
- (5) What is the difference in the number of eggs sold in last two days?

(B) The following graph shows the number of animals in the zoo.

Here each picture represents **2** animals.

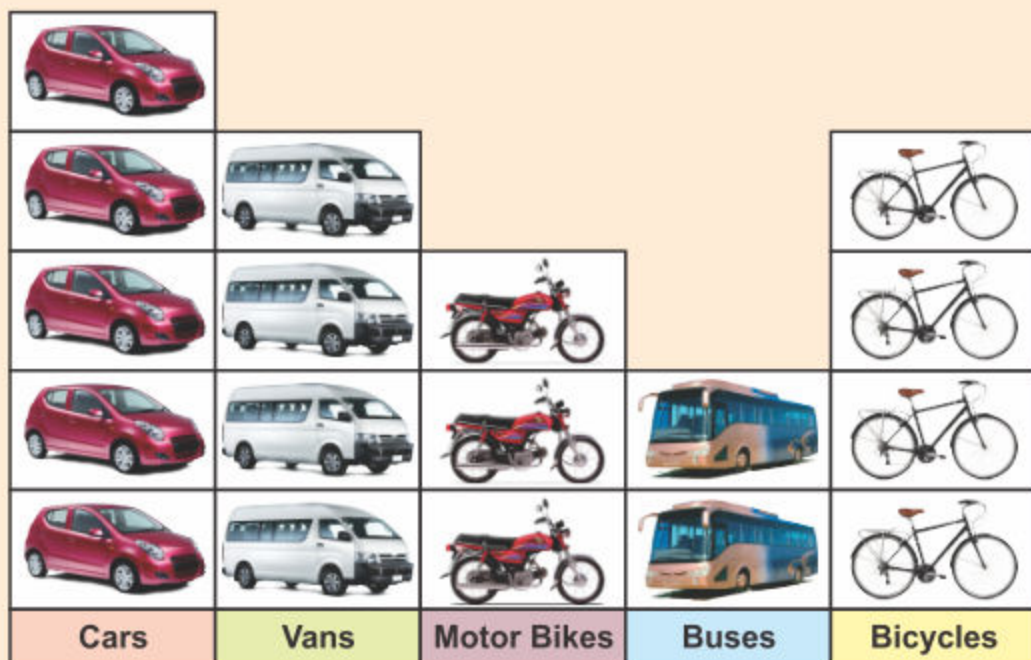


Look at the picture graph and answer the following:

- How many lions are there?
- How many animals are there altogether?
- How many more foxes are there than elephants?
- How many different types of animals are there?
- Which animals are most in number?
- Which animals are least in number?

- (C) Following is picture graph of the traffic passing through a signal.

One picture = 10 vehicles



Now look at the graph and answer the following questions:

- (1) Which vehicles are more in numbers?
- (2) Which vehicle are least in numbers?
- (3) How many buses passed through the signals?
- (4) How many motor bikes passed through the signals?
- (5) How many vans and bicycles passed altogether?
- (6) How many more cars passed than the buses?

GLOSSARY

Addition:	Symbol +; the process of finding sum of two numbers/quantities.
Capacity:	The amount of liquid a container can hold.
Centimetre:	A unit of length, 100 centimetres (cm) = 1 metre (m)
Circle:	A plane shape bounded by a single curved line where all of its points are at equal distance from a fixed point.
Currency:	Money in any form.
Division:	Process of finding quotient of two number/quantities. (The repeated subtraction).
Denominator:	Lower number of the common fraction.
Diameter:	A half circle's line segment is called diameter of the circle 
Even numbers:	The numbers having 0,2,4,6,8 at their units place.
Edge:	A one dimensional line segment joining two vertices.
Fraction:	Equal parts of a whole.
Gram:	Unit of mass.
Graph:	A pictorial representation of a data.
Hours:	24 th part of the day, 60 minutes. A unit of time 1 hour = 60 minutes
Improper fraction:	A fraction whose numerator is greater than the denominator.
Kilogram:	A unit of mass. 1 kilogram (kg) = 1000 grams (g)
Litre:	Unit of volume/capacity. 1 litre (l) = 1000 millilitres (ml)
Line segment:	Shortest distance between two points. 
Line:	 This figure represents a line AB.
Lunar Calendar:	(Hijrah Qamri Calendar) Islamic Calendar in a solar year.
Multiplication:	The process of finding product of two numbers/quantities (Repeated Addition).
Mass:	Quantity of matter present in a body.
Millilitre:	Thousandth part of a litre.
Millimetre:	Thousandths part of a metre.

GLOSSARY

Minute:	Sixtieth part of an hour. 1 minute = 60 seconds
Month:	A unit of time. 1 month = 30 days
Numerator:	Upper number of common fraction.
Odd numbers:	The numbers having 1, 3, 5, 7, 9 at their units place.
Place value:	Value of a digit of a number according to its place.
Proper fraction:	A fraction whose numerator is less than the denominator.
Paisa:	Unit of Pakistani currency.
Point:	A small dot used for location of a place on any surface.
Quadrilateral:	A four sided closed figure.
Remainder:	The number left over when one integer is divided by another.
Ray:	An arrow mark on one end point of a line segment  A Ray AB B
Rectangle:	A quadrilateral whose opposite sides are equal and have four right angles.
Radius:	The distance from the centre of the circle to the boundary of the circle.
Rupee:	Unit of Pakistani currency.
Ruler:	A straight edge used to measure distances.
Symbol:	A sign used to represent an operation, element or relation.
Square:	A quadrilateral whose all four sides are equal and has four right angles.
Subtraction:	Symbol (-). The process of finding the difference between two numbers/quantities.
Second:	Unit of time, $\frac{1}{60}$ the part of a minute.
Solar Calendar:	In this calendar, the dates indicates the position around the sun (365 days in a year).
Triangle:	A three sided closed figure.
Unit:	A list or sequence with only one element.
Vertex:	An angular point of any shape.
Week:	A unit of time. 1 week = 7 days
Year:	A unit of time. 1 year = 365 days

ANSWERS

EXERCISE 1

- (1) V, VII, XI, IX, VIII, XVI, XVII, XX
- (2) 3, 4, 6, 10, 2, 15, 12, 19
- (3) II, IV, V, VII, VIII, IX, XI, XII, XIV, XV, XVI, XVIII, XIX, XX
- (6) I, II, III, IV, V, VI, VII, VIII, IX, X, XI, XII, XIII, XIV, XV, XVI, XVII, XVIII, XIX, XX

EXERCISE 2

- (3) 2, 4, 6, 8, 10, 12, 14, 16, 18
- (4) 3, 5, 7, 9, 11, 13, 15, 17, 19
- (5) 32, 34, 36, 38, 40 (6) 51, 53, 55, 57
- (7) 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 78, 80
- (8) 71, 73, 75, 77, 79, 81, 83, 85, 87, 89
- (9) i. False ii. False iii. True iv. True

EXERCISE 4

- (1) i. Four thousand one hundred fifty six
ii. Thirty five thousand six hundred seventy eight
iii. Nine hundred eighty one thousand two hundred fifty four
iv. Five thousand sixty
v. Seven thousand one hundred twenty
vi. Three hundred fifty six thousand seven hundred eighty two
vii. Forty one thousand five hundred twenty four
viii. One hundred twenty three thousand four hundred fifty six
- (2) i. 8461 ii. 72695 iii. 55001
- (3) i. 950 ii. 25646 iii. 5421
- (4) i. 5124, 6124, 7124 ii. 51457, 51467, 51477
iii. 423456, 523456, 623456

EXERCISE 5

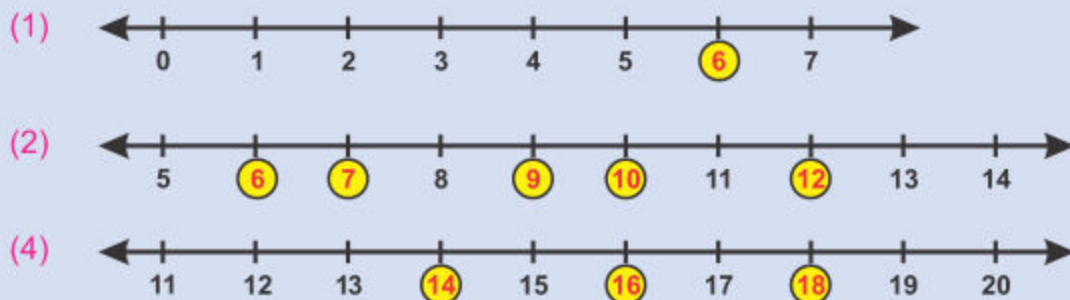
- (1) i. < ii. > iii. = iv. > v. < vi. >
- (2) i. > ii. > iii. = iv. < v. > vi. <

ANSWERS

EXERCISE 6

- (1) i. 476, 5034, 7881, 523241 ii. 1778, 2346, 2354, 4632
iii. 3451, 41353, 43513, 53314
- (2) i. 7120, 6432, 4343, 3213 ii. 94321, 49321, 31249, 12349
iii. 120451, 57401, 12345, 10000

EXERCISE 7



EXERCISE 8

1. 67 2. 455 3. 89 4. 199 5. 719
6. 1087 7. 1020 8. 923 9. 813 10. 49
11. 185 12. 938 13. 749 14. 769 15. 500

EXERCISE 9

- (A) 1. 3885 2. 4795 3. 9778 4. 7796 5. 9897
6. 5999 7. 7969 8. 9897 9. 8899
- (B) 1. 3537 2. 7535 3. 5859 4. 7796
5. 9897 6. 9839 7. 3765

EXERCISE 10

- (A) 1. 8020 2. 8071 3. 10183
4. 5891 5. 8706 6. 6659
- (B) 1. 7020 2. 5692 3. 12350 4. 7431
- (C) 1. 5083 2. 4724 3. 6504 4. 6432

ANSWERS

EXERCISE 11

- (A) 1. 19 2. 21 3. 24 4. 32
5. 28 6. 43
- (B) 1. 45 2. 43 3. 60 4. 36
5. 70 6. 80

EXERCISE 12

1. 1362 pages 2. 2200 m 3. 7986 population
4. 5793 visitors 5. 378 pages 6. 777 students
7. 8000 soft drinks

EXERCISE 13

1. 41 2. 22 3. 601 4. 322
5. 322 6. 322 7. 223 8. 147
9. 59 10. 513 11. 759 12. 187

EXERCISE 14

- (A) 1. 1237 2. 7234 3. 4442 4. 5522 5. 2212
6. 4633 7. 3402 8. 1533 9. 1123
- (B) 1. 2112 2. 3411 3. 2344 4. 4322 5. 4222

EXERCISE 15

- (A) 1. 2168 2. 3361 3. 1436 4. 1205 5. 2771
6. 3969 7. 1358 8. 1124 9. 736
- (B) 1. 5113 2. 844 3. 991 4. 4177
- (C) 1. 4039 2. 1771 3. 4028 4. 3344

EXERCISE 16

- (1) i. 9 ii. 62 iii. 66 iv. 63 v. 21 vi. 66
(2) i. 31 ii. 6 iii. 38 iv. 25 v. 49 vi. 53
vii. 79 viii. 13 (3) i. 6 ii. 5 iii. 22

EXERCISE 17

1. 370 chocolates 2. Rs 500 3. 514 cars
4. Rs 2055 5. 2638 cookies 6. 4629

ANSWERS

EXERCISE 18

- (A) 2. 12 3. 9 4. 35 5. 45 6. 24
(B) 2. 16 3. 30

EXERCISE 19

- (A) 1. 28 2. 62 3. 84 4. 48 5. 69
6. 70 7. 66 8. 318 9. 332
(B) 1. 26 2. 256 3. 115 4. 132
5. 147 6. 45 7. 195 8. 84

EXERCISE 21

1. 20 2. 30 3. 42 4. 54 5. 108
6. 112 7. 143 8. 208 9. 621 10. 928
11. 1400 12. 4875 13. 1215 14. 952 15. 1659
16. 2975 17. 2300 18. 3800

EXERCISE 22

1. 90 motor bikes 2. 280 weeks 3. 48 birds
4. 378 chocolates 5. 144 scoops 6. 150 books

EXERCISE 23

- (A) 1. 7 2. 8 3. 8 4. 9 5. 3
6. 9 7. 12 8. 17 9. 19 10. 15
(B) 1. 5 2. 7 3. 8 4. 10 5. 9
6. 8 7. 15 8. 33 9. 13 10. 19
11. 13 12. 25
(C) 2. Quotient 3. Divisor 4. Quotient 5. Divisor 6. Dividend

EXERCISE 24

1. 7 2. 9 3. 7 4. 20 pages
5. 7 copies 6. 3 litres 7. 9 toffees

EXERCISE 25

1. 20 packets 2. 16 boxes 3. 3 teams
4. 7 boxes 5. 14 kg

ANSWERS

EXERCISE 26

- (A) 1. $\frac{1}{4}$ 2. $\frac{7}{10}$ 3. $\frac{2}{3}$ 4. $\frac{3}{7}$ 5. $\frac{5}{12}$ 6. $\frac{5}{7}$
 (C) 1. $\frac{4}{11}$ 2. $\frac{7}{8}$

EXERCISE 27

- (1) (i) $\frac{2}{4}$, $\frac{1}{3}$, $\frac{3}{6}$, $\frac{1}{2}$ and $\frac{2}{4} = \frac{3}{6} = \frac{1}{2}$
 (ii) $\frac{1}{2}$, $\frac{2}{4}$, $\frac{2}{6}$, $\frac{3}{6}$ and $\frac{1}{2} = \frac{2}{4} = \frac{3}{6}$
 (iii) $\frac{2}{5}$, $\frac{3}{6}$, $\frac{2}{4}$, $\frac{1}{2}$ and $\frac{3}{6} = \frac{2}{4} = \frac{1}{2}$
 (2) (i) $\frac{2}{8}$, $\frac{3}{12}$, $\frac{4}{16}$ (ii) $\frac{4}{6}$, $\frac{6}{9}$, $\frac{8}{12}$ (iii) $\frac{6}{8}$, $\frac{9}{12}$, $\frac{12}{16}$
 (iv) $\frac{8}{10}$, $\frac{12}{15}$, $\frac{16}{20}$ (v) $\frac{2}{10}$, $\frac{3}{15}$, $\frac{4}{20}$ (vi) $\frac{4}{12}$, $\frac{6}{18}$, $\frac{8}{24}$

EXERCISE 28

- (A) 1, 2, 5 are Proper fraction
 (B) 1. < 2. > 3. < 4. > 5. =
 6. > 7. < 8. > 9. =

EXERCISE 29

- (A) 1. $\frac{2}{3}$ 2. $\frac{4}{5}$ 3. $\frac{3}{4}$ 4. $\frac{5}{6}$ 5. $\frac{6}{7}$
 6. $\frac{7}{8}$ 7. $\frac{9}{9}$ 8. $\frac{9}{11}$ 9. $\frac{13}{15}$ 10. $\frac{17}{18}$
 11. $\frac{13}{14}$ 12. $\frac{19}{20}$
 (B) 1. $\frac{5}{7}$ 2. $\frac{9}{12}$ 3. $\frac{3}{15}$
 (C) 1. $\frac{3}{5}$ 2. $\frac{8}{7}$ 3. $\frac{9}{10}$ 4. $\frac{2}{12}$ 5. $\frac{9}{13}$ 6. $\frac{11}{15}$

EXERCISE 30

- (1) i. $\frac{1}{3}$ ii. $\frac{2}{5}$ iii. $\frac{1}{6}$ iv. $\frac{3}{8}$ v. $\frac{2}{7}$
 vi. $\frac{2}{9}$ vii. $\frac{5}{10}$ viii. $\frac{4}{13}$ ix. $\frac{5}{11}$ x. $\frac{7}{12}$
 xi. $\frac{1}{15}$ xii. $\frac{5}{18}$

ANSWERS

- (2) i. $\frac{1}{4}$ ii. $\frac{11}{14}$ iii. $\frac{1}{19}$
- (3) i. $\frac{2}{7}$ ii. $\frac{7}{9}$ iii. $\frac{1}{10}$ iv. $\frac{7}{13}$ v. $\frac{4}{11}$
- vi. $\frac{6}{17}$ vii. $\frac{6}{25}$ viii. $\frac{1}{19}$ ix. $\frac{3}{15}$

EXERCISE 31

- (1) ii. 3cm iii. 7cm iv. 10cm v. 8cm
- (2) ii. ✓ iii. ✓ iv. ✗ v. ✗ vi. ✓
- (3) i. centimetre ii. centimetre iii. Centimetre
iv. metre v. metre

EXERCISE 32

- (A) 1. 88cm 2. 122m 3. 43km 4. 12m 74cm
5. 14m 99cm 6. 47km 951m
- (B) 1. 89cm 2. 27cm 3. 41km 4. 122m 35cm

EXERCISE 33

1. 103cm 2. 345m 3. 335km 4. 1585m

EXERCISE 34

- (A) 1. 39cm 2. 29km 3. 18m
4. 22km 32m 5. 18km 22m 6. 29km 178m
- (B) 1. 40cm 2. 3m 3. 7km
4. 186m 5. 72km 174m

EXERCISE 35

1. 15m wire 2. 25cm lace 3. 634m ribbon 4. 2m 28cm

EXERCISE 36

- (1) (i) g (ii) kg (iii) kg (iv) g (v) g (vi) kg (vii) g (viii) kg
- (2) (ii) kg (iii) g (iv) kg (v) g (vi) g

ANSWERS

EXERCISE 37

- (A) (1) 595 g (2) 817 g (3) 802 g (4) 81 kg
(5) 3667 kg (6) 81221 kg
- (B) (1) 550 g (2) 1023 kg (3) 1275 g (4) 7797 g

EXERCISE 38

- (1) 600 g spices (2) 20 kg flour (3) 59 kg cotton (4) 15 kg vegetables

EXERCISE 39

- (A) (1) 1 kg (2) 114 kg (3) 159 g (4) 379 kg
(5) 111 kg (6) 589 g
- (B) (1) 252 g (2) 253 kg (3) 220 g (4) 8 kg
(5) 220 g (6) 80 kg

EXERCISE 40

- (1) 15 kg onions (2) 100 kg rice (3) 1450 kg iron (4) 22 kg more

EXERCISE 41

- (A) (1) 280 ml (2) 3 l (3) 500 ml
- (B) (2) 5 ml (3) 100 l (4) 50 ml (5) 200 ml

EXERCISE 42

- (A) (1) 858 ml (2) 1020 ml (3) 999 l (4) 1000 l (5) 1665 ml (6) 5 l 769 ml
- (B) (1) 495 ml (2) 1355 l (3) 5 l 990 ml (4) 59 l 825 ml

EXERCISE 43

- (1) 550 ml (2) 33 l (3) 3365 l (4) 925 l 900 ml

EXERCISE 44

- (A) (1) 125 ml (2) 461 l (3) 2174 l
(4) 241 ml (5) 88 ml (6) 15 ml
- (B) (1) 115 ml (2) 440 l (3) 180 ml (4) 110 l

EXERCISE 45

- (1) 330 ml oil (2) 42 l petrol (3) 15 l water (4) 650 l milk

ANSWERS

EXERCISE 46

- (1) (ii) 5:45 p.m. (iii) 3:00 p.m. (iv) 4:30 a.m.
(v) 5:55 p.m. (vi) 11:40 p.m. (vii) 2:30 p.m.
(viii) 1:45 a.m. (ix) 10:45 p.m.
- (2) (ii) 3:45 p.m. (iii) 10:30 p.m. (iv) 1:45 a.m.
(v) 6:30 a.m. (vi) 3:45 a.m.
- (3) (ii) Quarter past 7'o clock in the evening.
(iii) Half past 6'o clock in the morning.
(iv) Quarter to 10'o clock in the night.

EXERCISE 47

- (2) 10:35, 35 past 10 (3) 8:30, 30 past 8
(4) 1:15, 15 past 1 (5) 9 past 25 in the morning
(6) 11 past 30 in the morning (7) 3 past 40 in the afternoon
(8) 5 past 15 in the evening

EXERCISE 48

- (A) (1) Sunday (2) Thursday
(3) Monday (4) Saturday
(5) Sunday (6) Sunday
- (B) (1) Sunday (2) Thursday (3) 4
(5) January, March, May, July, August, October, December
(6) Wednesday

EXERCISE 49

- (A) (1) 7 hours (2) 29 hours (3) 41 hours
(4) 50 hours (5) 61 hours (6) 62 hours
- (B) (1) 54 hours (2) 28 hours (3) 225 hours
(4) 260 hours (5) 708 hours (6) 822 hours

EXERCISE 50

- (1) 17 hours (2) 121 hours (3) 7 hours (4) 9 hours

ANSWERS

EXERCISE 51

- (A) (1) 1 hour (2) 5 hours (3) 4 hours
(4) 15 hours (5) 19 hours (6) 10 hours
- (B) (1) 6 hours (2) 42 hours (3) 20 hours
(4) 105 hours (5) 237 hours (6) 417 hours

EXERCISE 52

- (1) Car is faster, 2 hours (2) 7 hours (3) Azeem, 4 hours more
(4) B, 6 hours more (5) 2 hours, 30 minutes

EXERCISE 53

- (1) (b) (2) (a) M, N (b) P
(3) $\overline{PQ}$ or $\overline{QP}$ (4) G
(5) $\overline{AB}$, $\overline{AD}$, $\overline{AE}$, $\overline{AC}$ (6) $\overrightarrow{PQ}$, $\overrightarrow{QR}$, $\overrightarrow{PR}$

EXERCISE 54

- (1) (i) Triangle (ii) Square (iii) Rectangle
(3) (i) Square (ii) Triangle (iii) Rectangle
(iv) Square (v) Rectangle

EXERCISE 56

- (1) 18 cm (2) 16 cm (3) 15 cm (4) 152 cm (5) 250 cm

EXERCISE 57

- (A) (1) 55 (2) 1st day (3) 3rd day (4) 90 (5) 20
(B) (1) 4 (2) 36 (3) 4 (4) 6 (5) Rabbits (6) Elephant
(C) (1) Cars (2) Buses (3) 4 (4) 6 (5) 18 (6) 8